

Documentation for Software Design Project

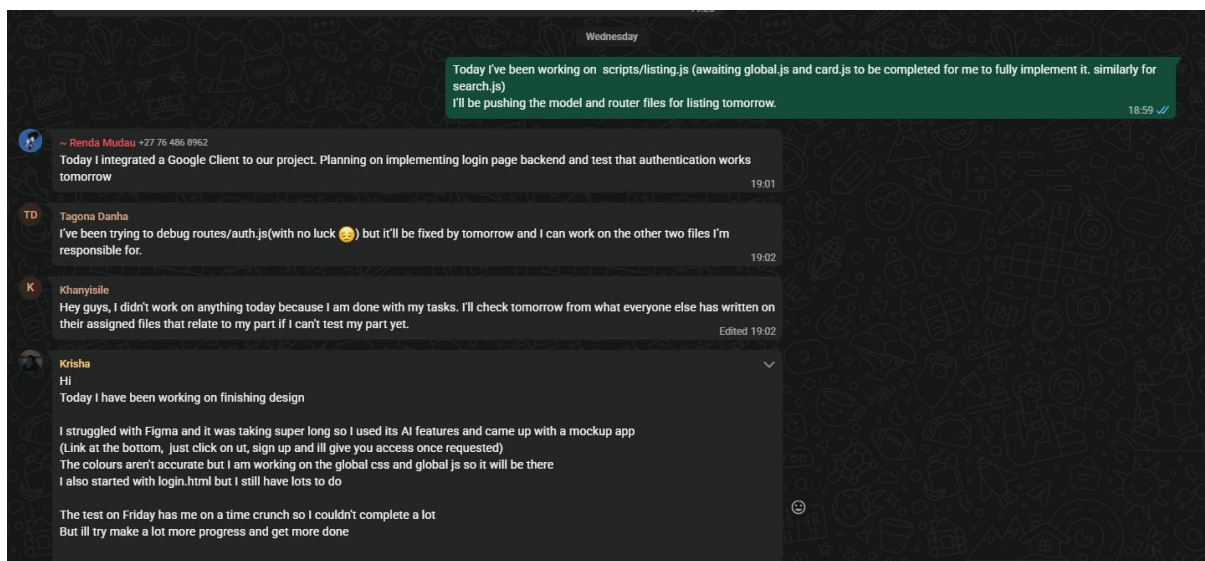
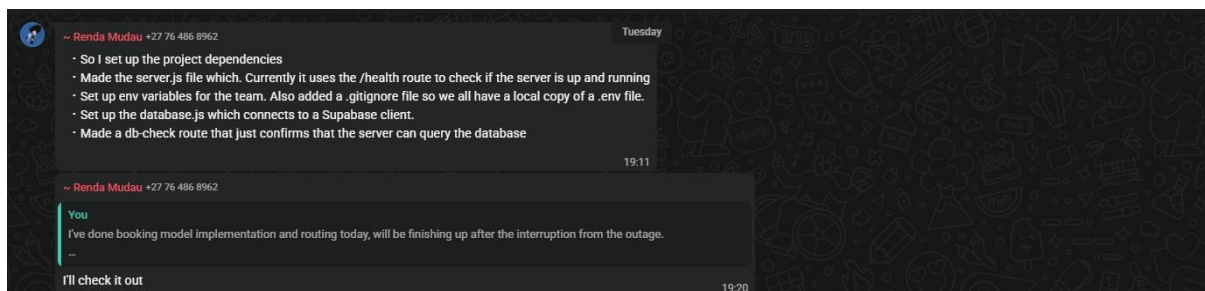
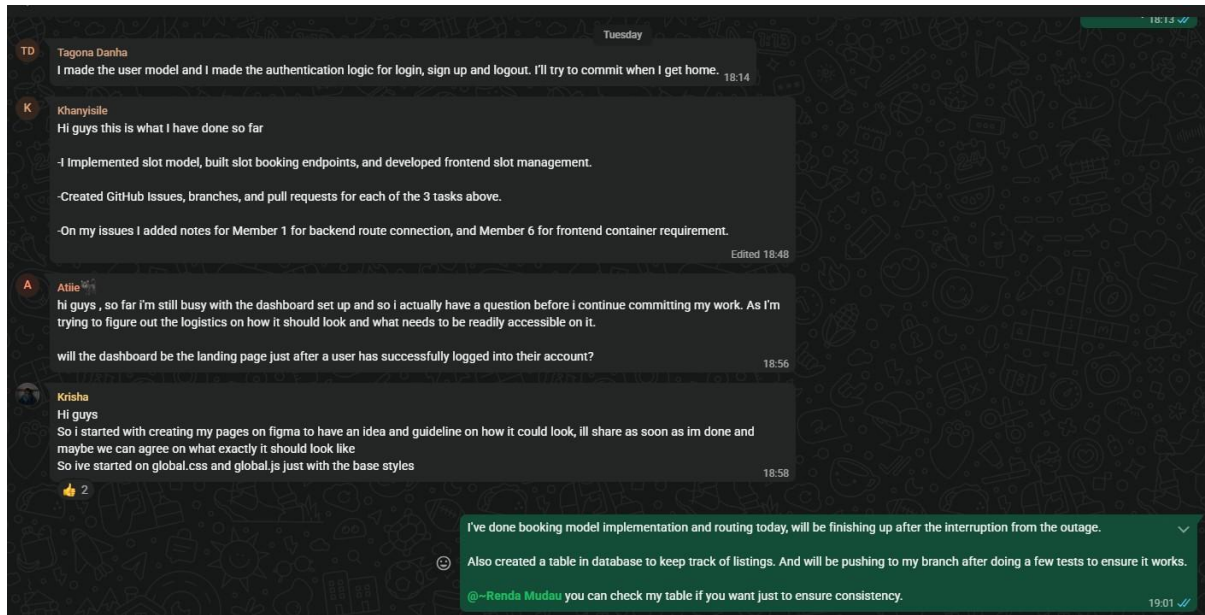
Compiled by: Zayden Suresh (2610978)

Contents

Documentation for Software Design Project	1
Sprint 1	2
Short WhatsApp meeting sessions:	2
Minutes for Meeting 2026/04/04	4
Minutes for Retrospective Meeting 2026/04/11	9
Sprint 2:	11
Short WhatsApp meeting sessions	11
Minutes for Meeting 15/04/2026	13
Sprint 3:	20
Sprint Planning Meeting Minutes – Sprint 3	20
Sprint 3 Retrospective (1)	21
Sprint 3 Retrospective (2)	24
Sprint 4	26
Sprint 4 Requirements Meeting	26
WhatsApp meetings	27
Sprint 4 Retrospective Meeting	30

Sprint 1

Short WhatsApp meeting sessions:



A

Atiie

Wednesday

earlier today i implemented a few things on the dashboard, such as the navigation bar etc.. but it's not all set and done as i still need to redo the trade slots page.

also wanted to see how the login page will look at the end and take inspo from it, in order to complete the look.

Also, is one responsible in implementing code for the files that fall under "Dependencies" in their respective task or someone else is responsible for them and we just use them once they're implemented?

19:14

TD

Tagona Danha

I added the user model, the routes/auth and i've written the scripts/auth

19:00

Thursday

Hey guys i added a line to server.js for routing listings, also added routing listings and model listings to the main repo Search.js and Scripts listings will be done once some other files are complete

18:59 ✓



~ Renda Mudau +27 76 486 8962

I just made the initial login routing for our app. This code was all added to routes/auth.js

I made /auth/google, /auth/logout and /auth/login. I plan on testing these endpoints tomorrow. Afterwards I can begin connecting OAuth to our project to track user sessions and keep cookies of their api and stuff.

19:05



Krishna

Hi guys

I made progress with global files, I should be finishing up by tomorrow or so

Still working on some parts of login.css and html

But I should make a lot more progress tomorrow after the test 🙏

19:07

Minutes for Meeting 2026/04/04

Discussions

~11:00-11:20

First 20 minutes:

Basic greetings complete, everyone has attended

Main objectives are making a product backlog, selecting user stories for sprint 1 and assigning tasks

~11:20-11:50

Confirmed most requirements, long discussion on the meaning of "facilities". Question to tutor for clarification

~11:50-12:20

Identifying the major features, additionally adding them to the product backlog.

Must user stories be added to the backlog or must it be features? For now, features, will ask tutor later for clarification

~12:20-13:00

Product backlog selection was decided upon: Student's ability to login, student creating item listing to sell, student searching for specific items, student wanting to book a drop-off time slot to complete trades.

These are now added to Trello

Each member broke down the user stories into tasks, we have selected appropriate tasks for user stories

BREAK (13:00-13:30)

~13:30-14:15

Been working on determining dependencies of each task for optimal task allocation. Split tasks into 6 categories (for each member to select which one they want to work on)

I had split the tasks as the following (Used an LLM to assist in determining dependencies):

Member 1 – Core Backend (Server & Database Setup)

Tasks package.json

Purpose: Define backend dependencies and scripts Responsibilities:

- Add required dependencies (e.g., Express, CORS, dotenv)
- Define scripts (start, dev)

Dependencies: None database.js

Purpose: Establish database connection

Responsibilities:

- Configure connection (MongoDB/MySQL)
- Handle connection errors • Export database instance Dependencies: None

server.js

Purpose: Main backend server setup

Responsibilities:

- Initialize Express server
- Configure middleware (JSON parser, CORS)
- Connect route files (auth, listings, slots)
- Start server

Dependencies: package.json, database.js

Potential Blockers & Solutions

- May be blocked if routes are incomplete • Solution: Use placeholder routes so server runs early

Member 2 – Authentication & Session Management

Tasks models/user.js Purpose:

Define user schema

Responsibilities:

- Store email and password
- Enforce unique email constraint

Dependencies: database.js routes/auth.js

Purpose: Authentication endpoints

Responsibilities:

- Implement login, registration, and logout
- Validate input and handle errors

Dependencies: server.js, models/user.js scripts/auth.js

Purpose: Frontend authentication logic

Responsibilities:

- Handle login/register form submission
- Send API requests to backend
- Process responses

Dependencies: Backend auth routes,

global.js scripts/session.js

Purpose: Manage user session

Responsibilities:

- Store session data (e.g., JWT) in localStorage • Check login status on page load
- Handle logout functionality

Dependencies: auth.js, global.js

Potential Blockers & Solutions

- May depend on backend being ready • Solution: Use mock API responses during development

Member 3 – Listings Management

Tasks models/listing.js Purpose:

Define listing schema

Responsibilities:

- Store title, description, price, and userId

Dependencies: database.js, models/user.js

routes/listings.js Purpose: Listings

API endpoints Responsibilities:

- Implement CRUD operations
- Associate listings with users

Dependencies: server.js, models/listing.js

scripts/listings.js Purpose:

Frontend listings logic

Responsibilities:

- Fetch and display listings
- Update and delete listings
- Implement filtering Dependencies: Backend routes, global.js,

card.js components/search.js Purpose:

Search and filtering component

Responsibilities:

- Filter listings dynamically
- Update UI results

Dependencies: scripts/listings.js, global.js

Potential Blockers & Solutions

- May depend on authentication system •

Solution: Use temporary user data during development

Member 4 – Slots / Trade Scheduling

Tasks models/slot.js Purpose: Define

slot schema Responsibilities:

- Store time, availability, and booking status
Dependencies: database.js routes/slots.js
Purpose: Slot booking endpoints
Responsibilities:

- Retrieve available slots

- Book slots

- Prevent double booking Dependencies: server.js, models/slot.js scripts/slots.js

Purpose: Frontend slot management

Responsibilities:

- Fetch and display slots

- Handle booking actions Dependencies:

Backend routes, global.js,

card.js

Potential Blockers & Solutions

- May depend on database setup
- Solution: Use hardcoded slot data initially

Member 5 – Frontend Login & Listing

Creation Tasks pages/login.html

Purpose: Login/register interface

Responsibilities:

- Create UI form

- Validate user input

- Display errors Dependencies: auth.js, global.css pages/create-listing.html

Purpose: Listing creation interface

Responsibilities:

- Build form for listing creation

- Validate input

- Submit data to backend Dependencies:

listings.js, auth.js, form.js, button.js

styles/login.css Purpose: Login page styling

Dependencies: global.css styles/listing.css

Purpose: Listing page styling Dependencies:

global.css scripts/global.js Purpose: Shared helper functions Responsibilities:

- Create reusable API request functions

(fetch wrapper)

- Handle global error handling

- Attach authentication tokens to requests

- Provide shared utility functions

Dependencies: Used by all frontend scripts

styles/global.css Purpose: Base styling

Responsibilities:

- Define global layout (body, containers)

- Standardize fonts, colors, and spacing •
Create reusable classes (buttons, forms, cards)

- Ensure consistent UI across all pages

Dependencies: Used by all pages

Potential Blockers & Solutions

- May depend on backend and scripts •

Solution: Use placeholder handlers for UI testing

Member 6 – Dashboard & Trade Slots

Frontend Tasks

pages/dashboard.html

Purpose: Main dashboard

Responsibilities:

- Display user info

- Show listings overview

- Include navigation Dependencies: listings.js, navbar.js, global.css pages/trade-slots.html

Purpose: Slot booking interface

Responsibilities:

- Display available slots

- Enable booking interactions Dependencies: slots.js, card.js, slots.css styles/slots.css

Purpose: Slot page styling

Dependencies: global.css

Potential Blockers & Solutions

- May depend on listings and slots scripts •

Solution: Use mock data during development

Shared Resources (Used by All Members)

Components

- components/navbar.js – Navigation bar

- components/card.js – Reusable display card

- components/form.js – Form wrapper with validation

- components/button.js – Reusable button

Global Files scripts/global.js – Shared helper

functions styles/global.css – Base styling

Renda - 1
Khanyisile - 4
TG - 2
Krisha - 5
Arnold - 6
Zayden - 3

Members have agreed tasks are reasonable

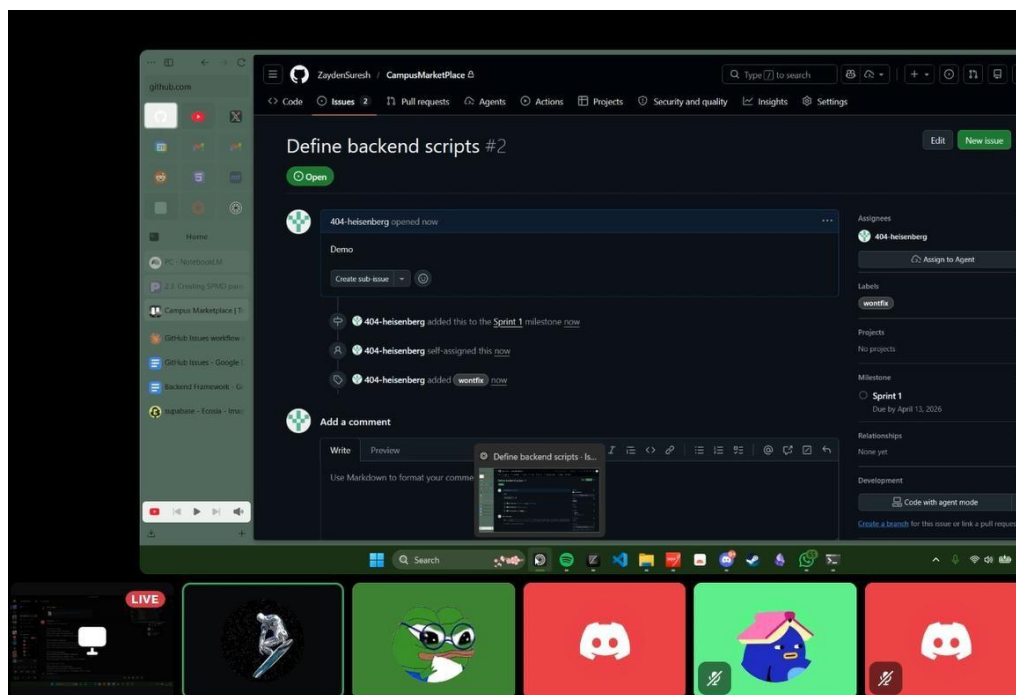
~14:15-14:30

UATs

Where will we submit UATs, when do we submit them? To be discussed with tutor.

~14:30-15:00

Will use <https://github.com/ZaydenSuresh/CampusMarketPlace> for project, Renda shows us how issues and branches work.



~15:00-15:10

Giving basic definition of Jest

Decided jest will be discussed more in sprint 2

CI/CD setup and Azure will also be discussed in sprint 2

~15:10-15:30

Decided we will log tasks on Trello, say what we've done on WhatsApp from Tuesday – Thursday.
Have a short online meeting on Friday

Team had questions on how tasks can be done when reliant on others (Communicate with member you are reliant on)

Decisions on task assignments reiterated, all members satisfied with their task allocation

Everyone expected to submit small bits of work throughout the week

Short closing for meeting complete

Minutes for Retrospective Meeting 2026/04/11



~10:00-10:30

Short greetings and fixing technical difficulties

Khanyisile pardoned due to medical reasons

~10:30-11:15

Each member reiterated tasks completed

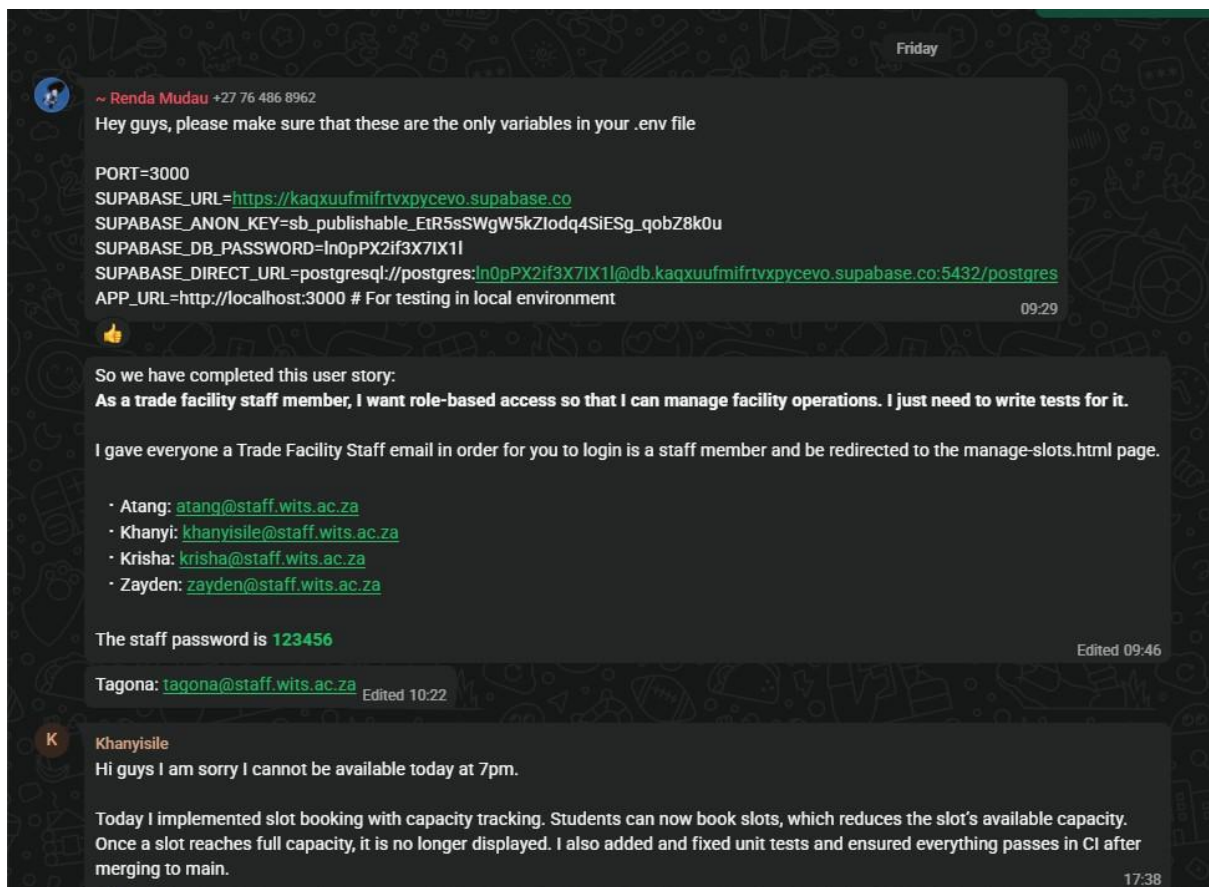
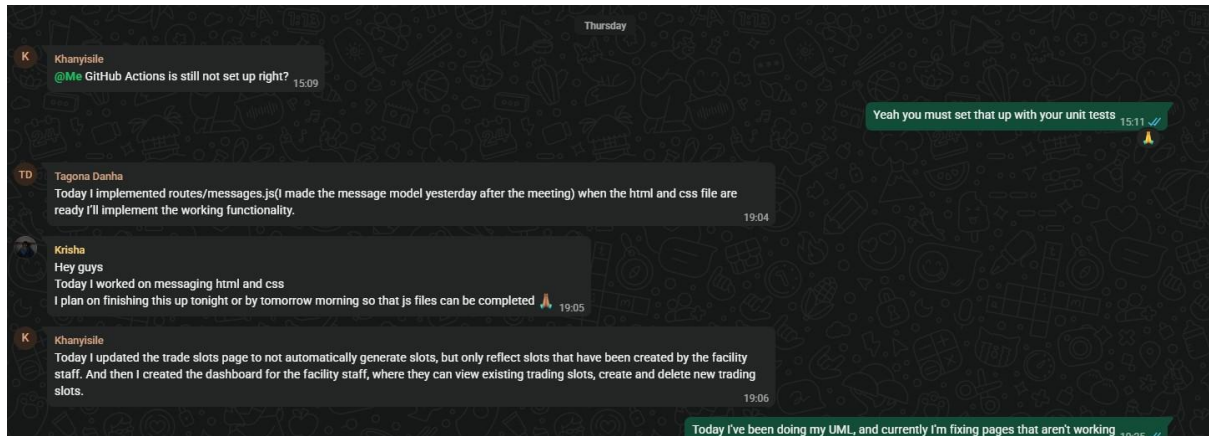
Still a few tasks needed to be completed, pushed retrospective meeting to Sunday instead

Determined inability to discuss completion of UATs and user stories due to tasks remaining

Discussed suggestions for better planning

Sprint 2:

Short WhatsApp meeting sessions



 ~ Renda Mudau +27 76 486 8962
Hi guys, I added my UML diagram to the repo under docs. I also finished all my tasks and just need to write the tests for them.

1. Added user role authentication. If you are a staff member you'll be taken to manage-slots.html upon sign in
2. Pages now check the role of the user when initiated. So a user who isn't a staff member can't go to manage-slots.html page and a user who isn't a student can't go to the dashboard.html page
3. I made the database filter listings rather than the app itself. Previously the database would return all listings and the app would filter them but now the listings are filtered by the database before and the app just renders them.
4. I updated scripts/auth.js to have reusable functions like getCurrentUser, logoutUser, registerUser etc.

Edited 19:05

I'm happy to join another team if y'all need the help. I just need to write tests for what I did 19:06

Hey guys today I just did documenting of our meeting on Wednesday, I made it so that you can upload images instead of adding links to stuff on the Internet. Also added an edit button to dashboard 19:09 ✓

 Krishna
Helloz
I finished the UI for messaging, all that needs to be done is the js which is currently in the works 🙌 Edited 19:10

I'll be working on my test functions mainly tomorrow, then I'll try and assist in the messaging aspect 19:11 ✓

TD Tagona Danha

 Krishna
Helloz
I finished the UI for messaging, all that needs to be done is the js which is currently in the works 🙌
I'm working on this. I think I'll be able to finish it by tomorrow afternoon. 19:11

A Attie 📱
hey guys, I'm currently still working on my UML diagram and thought the night I will be working on implementing the filter controls on dashboard.html and listings.css 19:19

TD Tagona Danha

Yesterday

~ Renda Mudau +27 76 486 8962
Guys are we not meeting today? We said we'd try to get all our code done by tonight
Hey, I've made progress on the messages. 19:22

K Khanyisile

~ Renda Mudau +27 76 486 8962
Guys are we not meeting today? We said we'd try to get all our code done by tonight
I think we should. 19:22

I have completed the 4th user story:

As a staff member, I want to manage booking slots so that facility capacity is controlled.

These are the headings of the user stories that I was following to implement :

- Create a slot
- Validate slot creation
- View all slots
- Edit a slot
- Prevent invalid capacity update
- Delete a slot
- Capacity decreases when booked (control capacity)
- Slot becomes unavailable when full
- Staff can monitor capacity in real time
- Prevent overbooking

20:34

To test this, you can log in using the email address that Renda sent. 20:35

Minutes for Meeting 15/04/2026

~18:10-18:20

Reviewed sprint 2 stories and updated backlog

~18:20-18:30

I briefly discussed what UML diagrams are and the 4+1 model

I asked a question to Renda on jest

~18:30-18:40

Decided to prepare better for viva and where to put UATs

~18:40-18:50

Better team coordination has been discussed

Renda has compiled the following:

Team 1: Auth & Listings Foundation

Main Features: User authentication, rolebased access control, and core listing stability

Dependencies: None (can start immediately) Tasks:

Stabilize email login

- Wire up email/password signup and login as proper API routes so students can log in without Google.
- Files: routes/[auth.js](#)

Add role field to user model

- Extend the user model to store role (Student, Admin, Facility Staff) and return it in auth responses.
- Files: models/[user.js](#)

Implement role-based middleware

- Create middleware to protect routes so only the right roles can access certain features.
- Files: routes/*.js

Store role in session

- Save the user's role in the client-side session so the frontend knows who is logged in.
- Files: scripts/[session.js](#)

Role-based UI elements

- Show/hide navigation items and

buttons based on the user's role. -
Files: components/navbar.js, pages/*.html

Backend search for listings

- Move search from client-side to a proper API endpoint so it works for all users.
- Files: routes/listings.js, components/[search.js](#)

Link slot bookings to users

- Associate each booking with the user who made it so students can view their appointments.
- Files: routes/slots.js, models/[slot.js](#)

Team 2: Discovery & Listing Management

Main Features: Enhanced listing experience with filters, edit functionality, and seller information

Dependencies: Team 1 (for session/role checks once edit permissions are added)

Tasks:

Listing edit/delete

- Allow listing owners to edit or remove their own listings. - Files: routes/listings.js, scripts/[listings.js](#)

Add filter UI

- Build filter controls for category, condition, and price range.
- Files: components/search.js, pages/dashboard.html, styles/listing.css

Send filter parameters to backend

- Connect frontend filters to the search API endpoint.
- Files: scripts/listings.js, routes/[listings.js](#)

Message Seller button

- Add a button on listings that opens the messaging system.
- Files: components/[card.js](#)

Display seller rating

- Show each seller's trust rating on their listings.
- Files: components/card.js

Team 3: Transactions & Communication

Main Features: In-app messaging and trust/rating system

Dependencies: Team 1 (for user identity)

Tasks:

Create messaging model

- Set up a database table to store message history between users.
- Files: models/[message.js](#)

Messaging API routes

- Create endpoints to send messages and retrieve conversation threads. - Files: routes/[messages.js](#)

Chat UI page

- Build a page where users can view and send messages.
- Files: pages/messages.html, scripts/messages.js, styles/messages.css

Create ratings model

- Set up database table to store user ratings after transactions.
- Files: models/[rating.js](#)

Ratings API routes

- Create endpoints to submit ratings and fetch a user's average rating. -
- Files: routes/[ratings.js](#)

~18:50-19:10

Brief discussion on fixing login page

Decided deadline for UML will be for Friday

Decided Renda will fix the login page

Khanyisile and Renda had a few questions

Decided on teams:

Team 1: Khanyisile and Renda

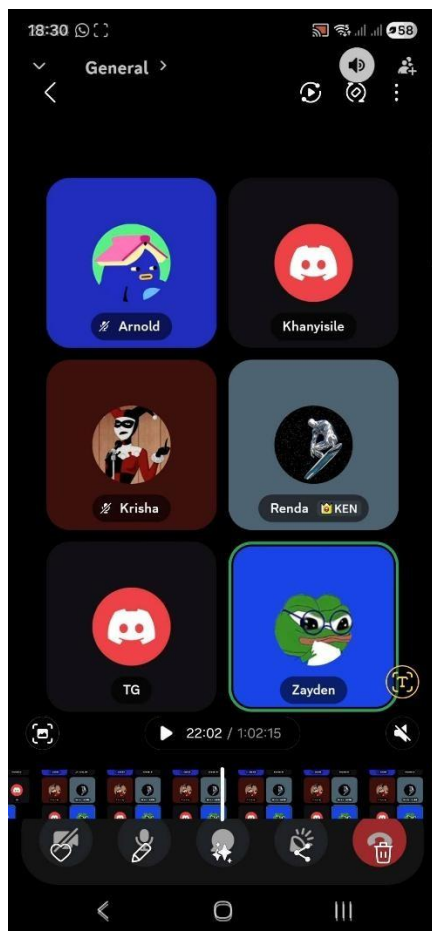
Team 2: Arnold and Zayden

Team 3: Tagona and Krisha

Meeting closed

Display ratings on dashboard

- Show seller trust scores on the main listings page.
- Files: scripts/ratings.js, pages/dashboard.html



Retrospective meeting (19/04/2026)

Written by Renda:

Sprint 2 Retrospective Meeting Minutes

1. Start and Greeting

The meeting was coordinated by Renda.

All group members were present, except for Zayden and Atang, who were absent. We greeted each other before proceeding with the session.

2. Show Work Completed

Each team member presented the work they completed during the sprint, demonstrating how their implemented user stories function within the application.

Members confirmed that:

- All assigned user stories had been completed
- Their work had been successfully pushed to GitHub
- Their features were functioning as expected when demonstrated

The group collectively reviewed the application and confirmed that the user stories for this sprint had been implemented successfully.

3. Reflect on the Sprint

3.1 Review of Previous Retrospective

The group revisited the Sprint 1 retrospective and identified that the main challenge previously was:

- Poor coordination due to members working independently on interdependent files

3.2 Improvements Implemented in This Sprint

To address this issue, the team adopted a new approach during Sprint 2 planning:

- Members were grouped into teams of two
- Each pair worked on related user stories with dependencies

3.3 Feedback on Improvements

Group members shared their experiences with this new approach:

- Collaboration within smaller teams improved coordination
- Communication between members became more effective
- Integration of dependent features was smoother

However, the group agreed that:

- Coordination can still be improved further
- Additional strategies may be needed to ensure better synchronization across all team members

4. Conclusion

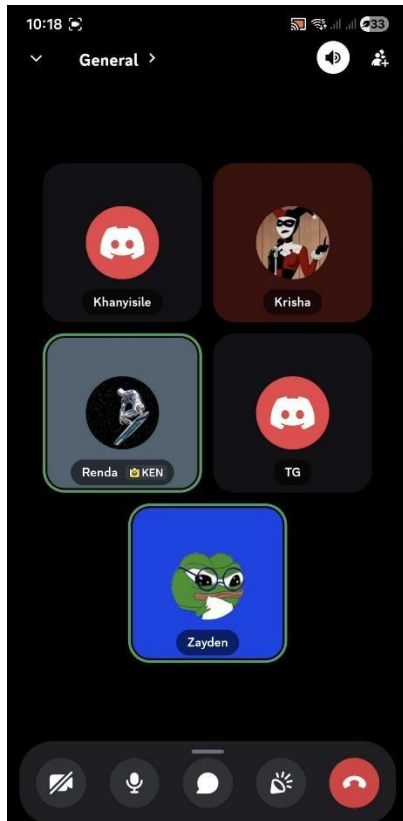
The team concluded that:

- Significant progress was made in improving coordination compared to Sprint 1
- The new team-based approach was beneficial
- Further improvements in communication and coordination will be explored in the next sprint

The meeting was then closed.

Sprint 3:

Sprint Planning Meeting Minutes – Sprint 3



Project: Campus Marketplace

Meeting Type: Sprint Planning

Date: 27 April 2026

Attendees: All team members attended

Next Meeting: 4 May 2026 (Sprint Refinement)

User Stories Selected

As a user (buyer or seller), I want to communicate through messaging before booking or completing a trade so that we can clarify and finalize transaction details. As an admin, I want to see popular item categories so that I understand trends.

As a user, I want to rate and leave a review for another user after a transaction so that I can share my experience and help build trust within the platform.

As a student buyer, I want to purchase or reserve an item so that I can initiate a transaction with the seller.

As a student, I want to delete my listings so that I can remove unavailable items.

As a staff member, I want to confirm both item drop-off and collection so that transactions are properly tracked and completed.

Additional Tasks Discussed

After selecting the user stories, the team broke them down into smaller development tasks and discussed fixes required in the system. The following issues were identified from the sprint 2 retrospective and review with the tutor:

Fix messages not properly displaying on the application.

Improve security for the messaging feature.

Replace unnecessary <div> usage with semantic HTML.

Fix the “My Listings” button functionality.

Remove the student view button from the staff dashboard because it incorrectly redirected staff users to the student dashboard.

Remove transaction analysis from the dashboard.

Remove reserved or purchased items from the dashboard listings.

Decisions Made

User stories were finalized for Sprint 3.

Tasks were allocated among the team members.

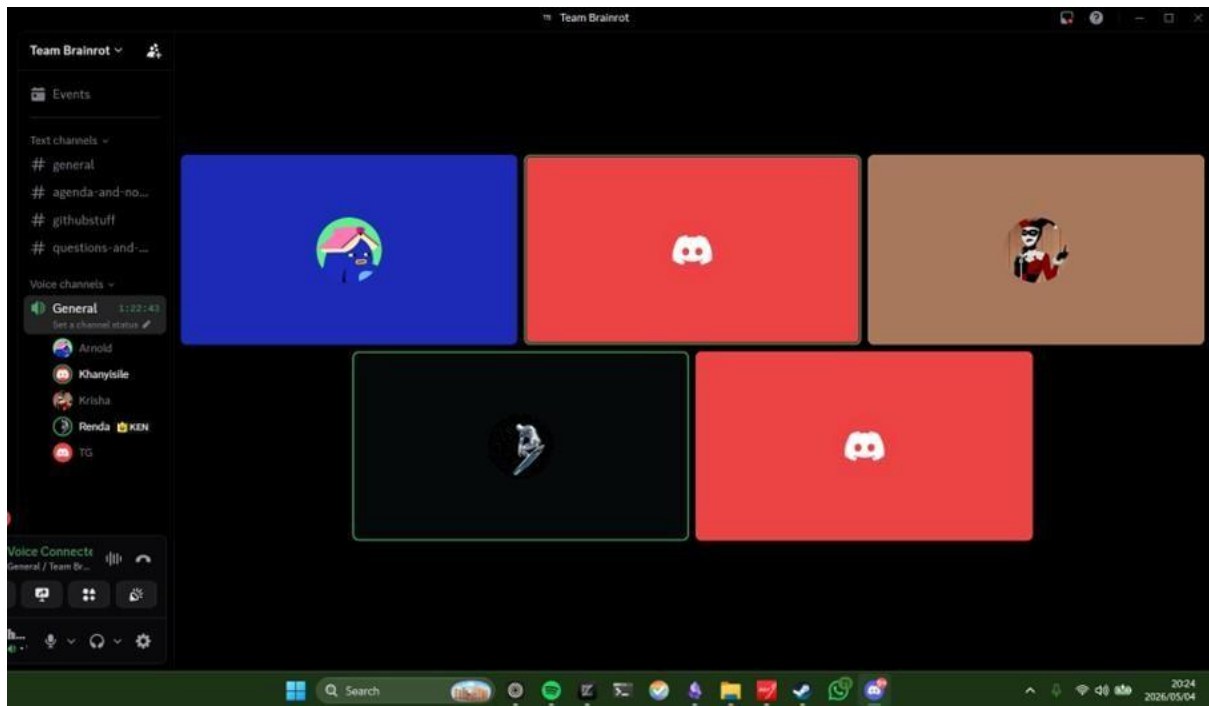
The next meeting on 4 May 2026 will focus on Sprint Refinement.

Sprint 3 Retrospective (1)

Meeting Minutes

Monday, 04 May 2025 · 19:00

Attendance



Atang	Krisha	Tagona	Renda	Khanyisile
-------	--------	--------	-------	------------

Sprint Updates

Summary of each team member's progress over the past two weeks.

Member	Status	Update / Notes
Renda	Completed	Delivered analytics dashboard (frontend + backend) covering all 16 transaction categories, with CSV export functionality.
Khanyisile	Completed*	Implemented Manage Trade Slots UI and full transaction workflow (Pending → Waiting Drop-Off/Pick-Up →
		Completed/Cancelled). Outstanding: separate transaction flow needed for Trades vs. Sales.

Atang	Blocked	Blocked pending Zayden's availability. Outstanding items: purchasing a listed item is non-functional; item deletion not yet started.
Tagona & Krisha	In Progress	Frontend for messaging and ratings is complete. Integration is blocked until Transactions are fully operational — ratings and meetings are inaccessible until a transaction exists in the system.

Application Workflow Discussion

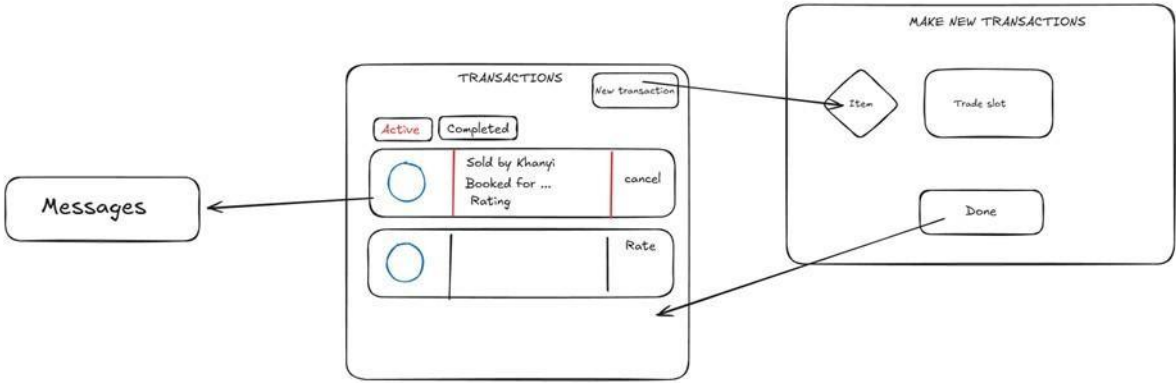
The team aligned on the end-to-end user journey from item reservation through to transaction completion.

Agreed Transaction Flow

- User reserves an item on the dashboard.
- User receives a notification to open a conversation with the seller.
- Buyer and seller negotiate and agree on a drop-off/collection time slot.
- Buyer creates a new transaction, selecting a reserved item and a trade slot.
- Trade facility staff tracks the transaction: Pending → In Progress → Completed.
- Once completed, the buyer can rate the seller.

Key Technical Decisions

New Transactions Page required — displays all transactions for a user (identical view for buyer and seller), showing current state. Allows cancellation of active transactions and rating of completed ones.



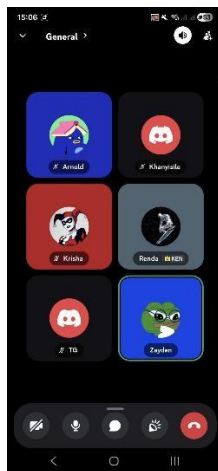
New reservedBy column to be added to the items database table — stores the buyer's user ID upon reservation. Only one user may reserve an item at a time.

Closing

The team reflected on the value of high-level application discussions for maintaining shared understanding. All members are aligned on the target state for the sprint and the steps needed to get there.

Sprint 3 Retrospective (2)

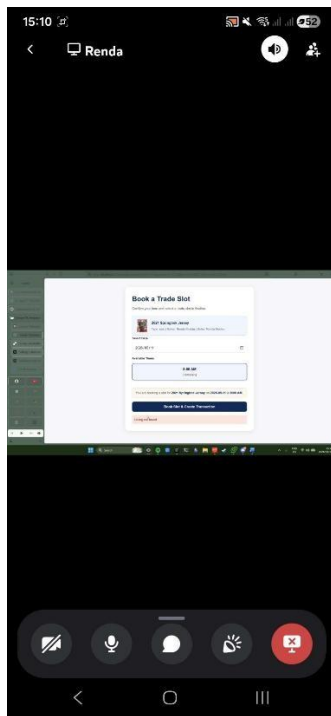
10 May 2026, 15:00 15:00-15:05:



Waiting for attendance, everyone is present

15:05-15:25

Each member discussed and presented what they did and what they have left to do

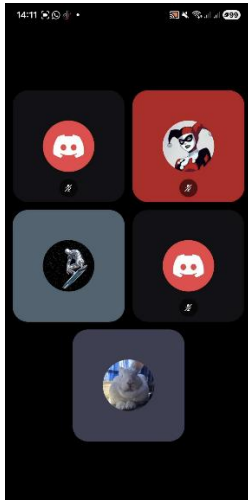


15:25:15:45

Discussed what needs to be fixed

Sprint 4

Sprint 4 Requirements Meeting



13 May 2026, 14:00-14:24

14:10-14:15:

Waiting for attendance. Team members joined the meeting and attendance was confirmed. Arnold could not make it due to an appointment.

14:15-14:20:

We had a discussion on the rubric and determined the importance of the final features, and that the project must be polished for submission

14:20-14:27:

We arranged tasks in order of a priority to satisfy most important rubric requirements:

SA Data Price Suggestion Integration
Online Payment and Cash Shortfall Tracking
Seller Transaction History Page
Admin Review Moderation System
Trade Facility Utilisation Analytics
Analytics Export and Moderated Content Report

14:27-14:31:

The team discussed the importance of unit tests being functional and working. The target is to reach 80% code coverage, I deleted my UATs document from the tests folder to ensure the code coverage is accurate.

14:31-14:33:

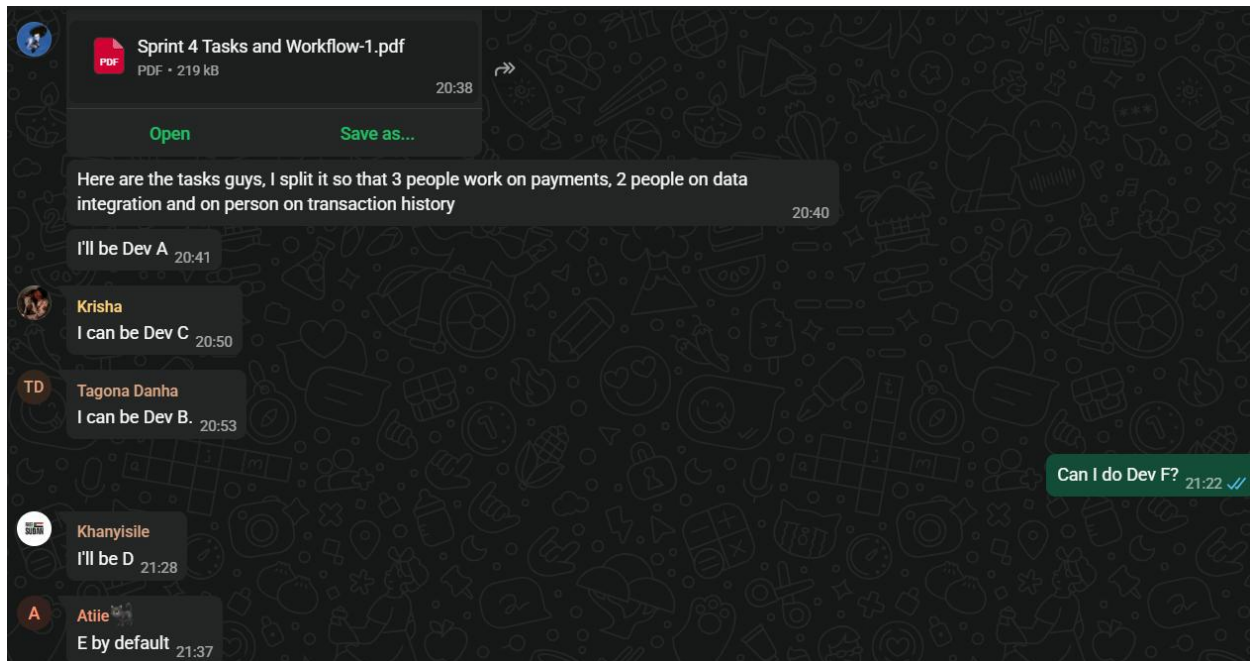
The team discussed final product polish. This included removing placeholder content, fixing visible bugs, checking that the UI is consistent, and making sure the system feels like a complete viable product before submission.

14:33-14:34:

The meeting closed with a short discussion of action items, and Renda will write the tasks which we will assign to ourselves later in the day

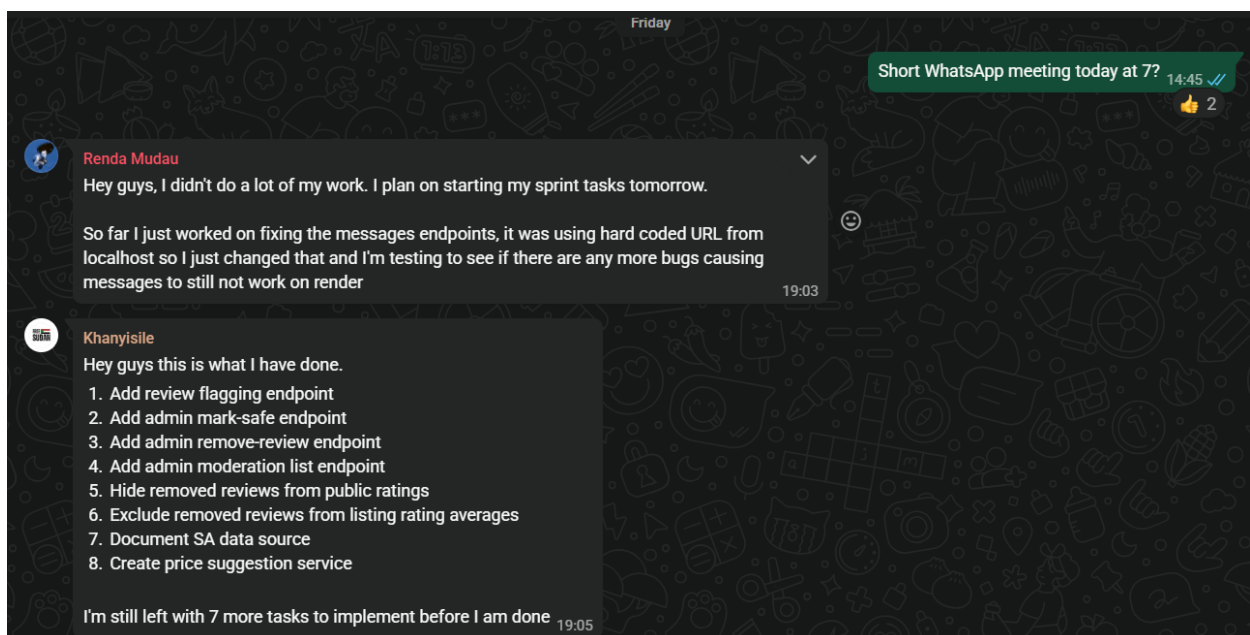
WhatsApp meetings

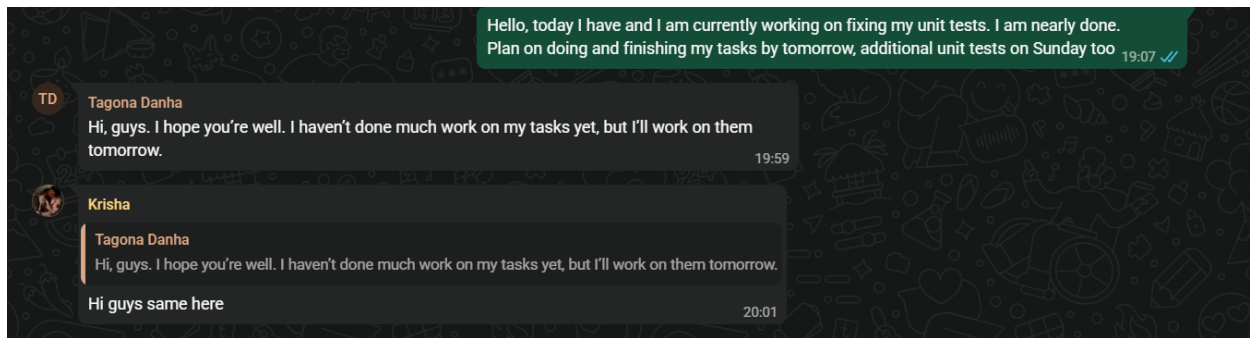
WhatsApp meeting (1) - Wednesday



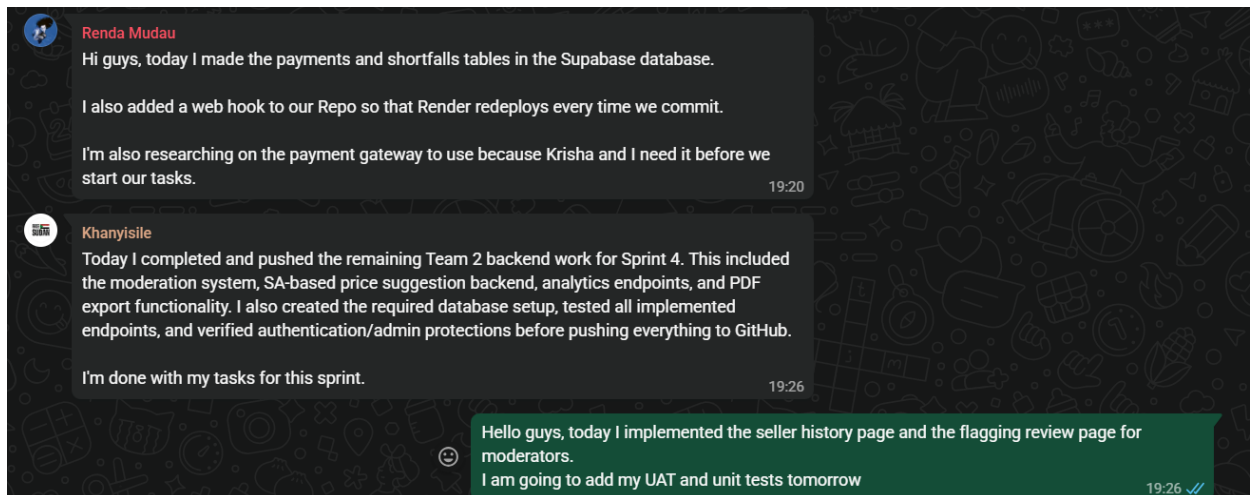
Task assignment was done (Refer to

WhatsApp meeting (2)





WhatsApp Meeting (3 - Saturday)





Khanyisile

Hey @Renda Mudau what was your idea of a suggested price? Which things were supposed to be considered before suggesting a price, was it just condition and category? Or there's something else to be considered

Anyone who has an idea please advise

21:12



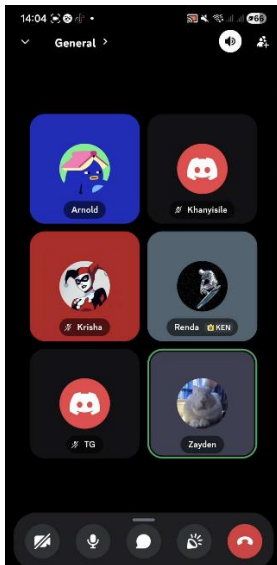
Atiie

hey guys

- i have added the export PDF button on the admin-dashboard.html page, just next to the pre-existing Export CSV button.
- I added the Slot Utilisation, Flagged Reviews & Removed Reviews stat cards at the top of the admin's dashboard.
- I added a Trade Facility Utilisation bar chart diagram at the bottom of the admin's dashboard, which renders with dates, using the prescribed dimensions and colour codes (Navy Blue Bar) for booked and a (Golden Bar) for Capacity. if there are no slot data available in the database, the bar chart shouldn't render and instead there should be a "No slot data available" line that appears in the space where the chart should sit.
- I implemented the price suggestion feature on the create-listing.html page, using the SA Market Data integration.

Edited 21:12

Sprint 4 Retrospective Meeting



~14:04-14:05 – Brief discussion on purpose of this final meeting

~14:05-14:10

Each member discussed what they had left

TG and Krisha still must finish their features

Krisha will push some of her code after the meeting

~14:12-14:17

Discussed that there are still a few unit tests left

Code coverage is shown as 87%

~14:17-14:22

Clarifications were made between a few members on their code

Renda willing to help other members with unit tests

We planned on checking the website, testing if there are any more small fixes that need to be made